
ZT-100 User Manual

Release Draft

ZONGE International

Sep 01, 2026

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TRANSMITTER SAFETY

Warning



The ZT100-B is a high power transmitter for geophysical purposes. It operates at lethal voltages. Only trained, qualified personnel should operate, maintain or service it.

- The ZT100-B is used together with a power supply which is typically fed by a generator. Make sure to comply with all safety rules and recommendations that come with these devices.
- Depending on country and industry the ZT100-B is used in, safety rules such as grounding and bonding requirements of the whole system may vary. Make sure to always comply with the relevant regulations for your survey.
- Before operating the ZT100-B make yourself familiar with all connections and with the operating instructions.

Note

Zonge recommends implementation of safe operations procedures for all transmission related operations.

ZT-100 CONTROLS AND CONNECTORS

2.1 Front Panel



Fig. 1: ZT100-B front panel.

From left to right the function of the ZT100-B front panel elements are:

1. The **Keyed ON/OFF** switch is connected to the ICE 320/C-14 connector in the back. It directly interrupts power to all internal boards. The loss of power on the internal electronics will also disengage the contactor for the main power input.
2. Pushing the **Emergency switch** will disengage the contactor interrupting the positive connection from the power supply to the ZT100-B. It will also put all switching elements into an open state. The power for the digital electronics will not be interrupted by the emergency switch.
3. Make sure the **Fans** are unobstructed at all times.
4. The **black mushroom antenna** above the Reset and Transmit buttons is the WiFi antenna of the ZT100-B. Make sure it is unobstructed for optimal performance.
5. Pushing the **Reset** button starts the reset/arm sequence, or stops transmission.
6. Pushing the **TRANSMIT** button starts transmission during the arm window.
7. The 128 x 64 pixel **transflective display** is readable in bright sunlight. A LED backlight allows operation at night.
8. Three push buttons **UP**, **DOWN**, and **SELECT (MODE)** are used for menu control.

2.2 Back Panel

From left to right the function of the ZT100-B back panel elements are:

1. **The four BNC monitoring outputs on the left have the following functions:**
 1. The **ON/OFF** monitor provides a digital, 3.3 V amplitude signal referencing the state of the ON/OFF control signal. This signal controls the flow of current. It is an active low signal, meaning the transmitter is in an ON state (current is flowing) when this signal is low.
 2. The **Polarity** monitor provides a digital, 3.3 V amplitude signal referencing the state of the polarity control signal. This signal controls the current direction from plus to minus or minus to plus.

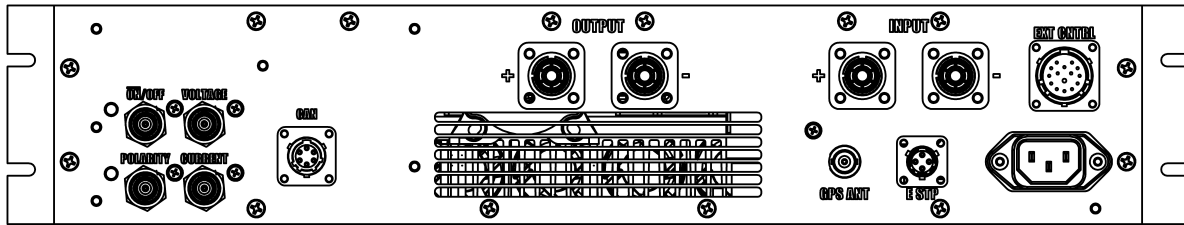


Fig. 2: ZT100-B back panel.

3. The **Voltage** monitor provides an isolated 2 mV/V representation of the output voltage.
4. The **Current** monitor provides an isolated 20 mV/A representation of the output current.
2. The **CAN Connector** next to the monitoring connectors connects to the internal CAN bus of the ZT100-B.
3. The **positive and negative output connectors** connect to the dipole or loop antenna used in a geophysical survey.
4. The **positive and negative input connectors** connect to the DC output of a suitable power supply. Always make sure to check the output polarity of the power supply matches the input polarity of the ZT100-B.
5. The **GPS antenna connector** is a 50 Ohm BNC connector for an active GPS antenna. The ZT100-B uses a GNSS module tracking GPS, Glonass and Baidu satellites. Use a 3.3 V active GPS antenna with minimum gain of 15 dB and maximum gain of 30 dB.
6. The **E STOP** connector allows to connect additional emergency stop switches.
7. The **External Control** connector allows to control the switching of the ZT100-B from an external controller.
8. The IEC 320/C-14 **Power Connector** is used to supply power for the internal electronics of the ZT100-B. The digital supply voltage must be between 85 VAC and 264 VAC.

FIRST USE

Note

Zonge recommends that new operators familiarize themselves with the equipment in a safe low voltage environment prior to using the equipment in a high voltage high current field situation.

3.1 Before You Start

This section describes a safe setup for the purpose of operator training without risks of a high voltage, high current environment. Such a setup is normally done in an laboratory environment. In addition to the ZT100-B you will need:

- A laboratory power supply capable to put out 10 V DC and 20 W. Ideally the power supply should have a voltage regulation and a current limiter.
- A low value high power resistor. A 4 Ohm or 10 Ohm resistor with at least 100 W will be sufficient.
- If available an oscilloscope or other kind of data-logger that can be used to display the monitoring signals.
- Cables and connectors

3.2 Connections

- In a geophysical survey the load is either a grounded dipole or a loop antenna. For this test a resistor is used as load. Connect the two ends of the resistor with the output cables. Make sure that the connection is solid and that the ends of the connector cables can not create a short. While this test will be run at low voltages current of 1 A or more will run through the wires. Bad connections can create a significant amount of heat.
- Connect the two input connectors cables to the output of the power supply. Make sure that the connections are solid and that the polarity is correct.
- Connect the GPS antenna delivered with the ZT100-B to the antenna connector of the ZT100-B. The ZT100-B GNSS unit is very sensitive and in many cases the synchronization will work with the antenna inside of a building, but ideally the antenna should be placed outside.
- Connect the **CURRENT** and **VOLTAGE** monitoring signals to an oscilloscope using 50 Ohm BNC cables. If you have access to a 4-channel oscilloscope connect the **ON/OFF** and **POLARITY** monitor signals as well.
- Connect the IEC 320/C-14 power connector to a household supply voltage between 85 VAC and 264 VAC.

3.3 Power On and Check Status

3.3.1 Power the ZT100-B On

Keep the laboratory power supply off for the moment and power on the ZT-100 by turning the ON/OFF key to the ON position. After a short showing of the Zonge logo the display should show:

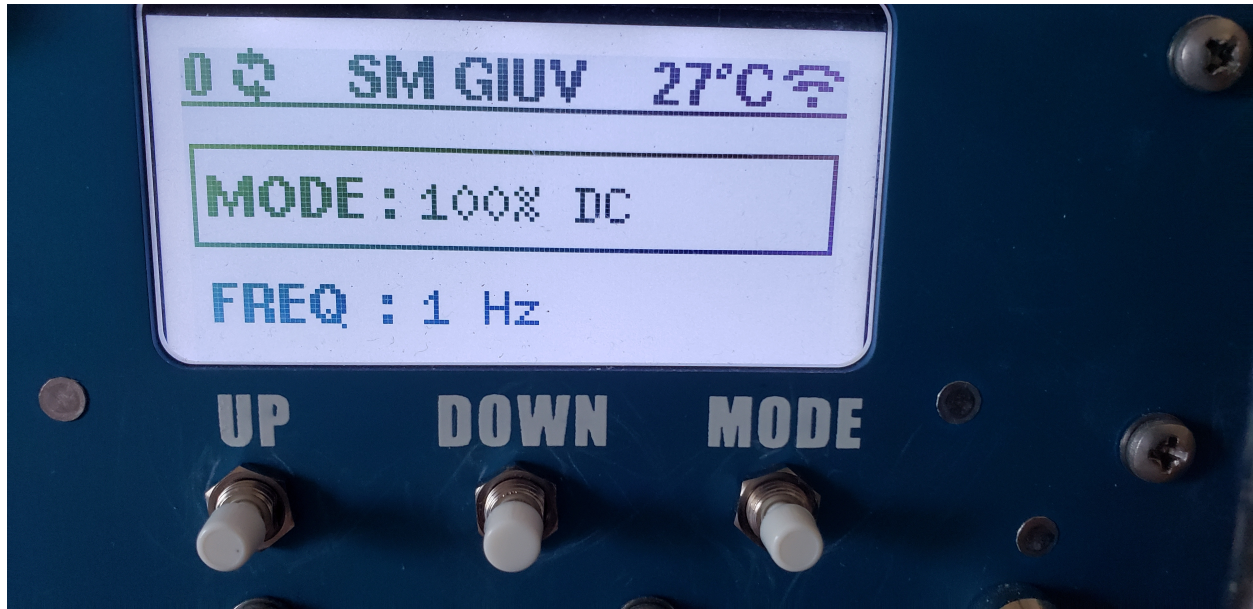


Fig. 1: ZT100-B display after turning on.

The number on the upper left shows how many GNSS satellite signal are currently received by the ZT100-B. Next to it are spinning arrow indicating that the timing of the ZT100-B is not locked to GNSS yet.

The middle section shows SM GIUV. This is an error message. SM stands for Safety Module and GIUV stands for General Input Under Voltage. The transmitter is complaining that it does not register any input voltage as the laboratory power supply is still turned off.

The temperature shown on the right is the internal temperature of the switching modules. This value will always be 25 deg C or higher even if the actual temperature is lower.

Finally the symbol on the right indicates that the WiFi mode of the ZT100-B is switched on.

3.3.2 Wait for the ZT100-B to lock to GNSS

Over the next couple of minutes the satellite count should increase. When enough GNSS satellite signals are received the ZT100-B will synchronize its internal timing with the PPS from the GNSS module. The timing is now locked to GNSS. This is indicated by a small padlock symbol replacing the spinning arrows.

3.4 Power Up the Laboratory Power Supply

Set the laboratory power supply to a voltage higher than 2 V. For convenience use a voltage that in combination with the resistor results in a 1 A signal. Make sure the current limiter of the power supply is set to a current of no more than 2 A.

After turning the power supply on press the **RESET** button of the ZT100-B. The transmitter will perform some tests and should show RESET in the middle of the upper display row.

3.5 Set Mode and Frequency

To change the operating mode (**100% DC**, **50%**, **Custom**, or **MMR 5Hz**) the **DOWN** button has to be pressed longer than a second. This is to prevent spontaneous change of mode during operation.

Note

In **MMR 5Hz** mode, frequency is fixed at 5 Hz and cannot be edited.

The frequency can be changed by pressing the **MODE** button shortly and then selecting the desired frequency with the **UP** and **DOWN** buttons.

3.6 Start Transmit

1. Press **RESET**.
2. You then have about 2 seconds to press **TRANSMIT**.
3. If everything is fine and load checks pass, transmit starts.

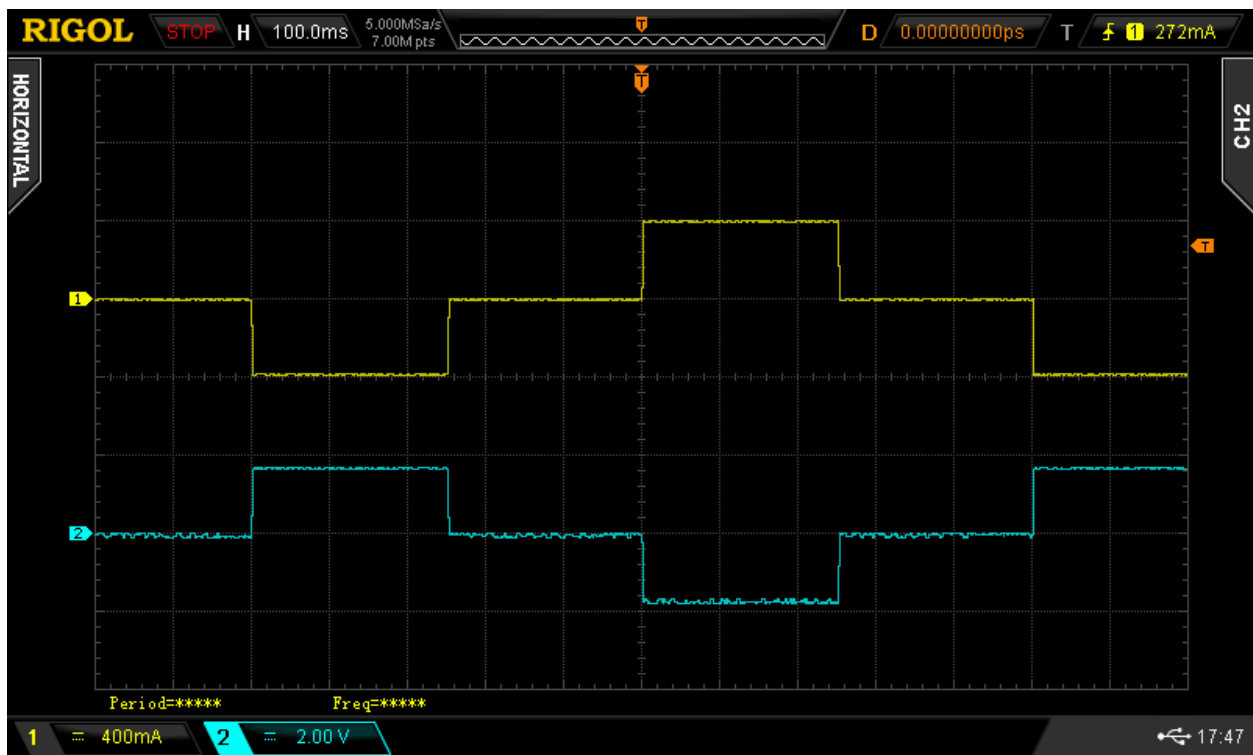


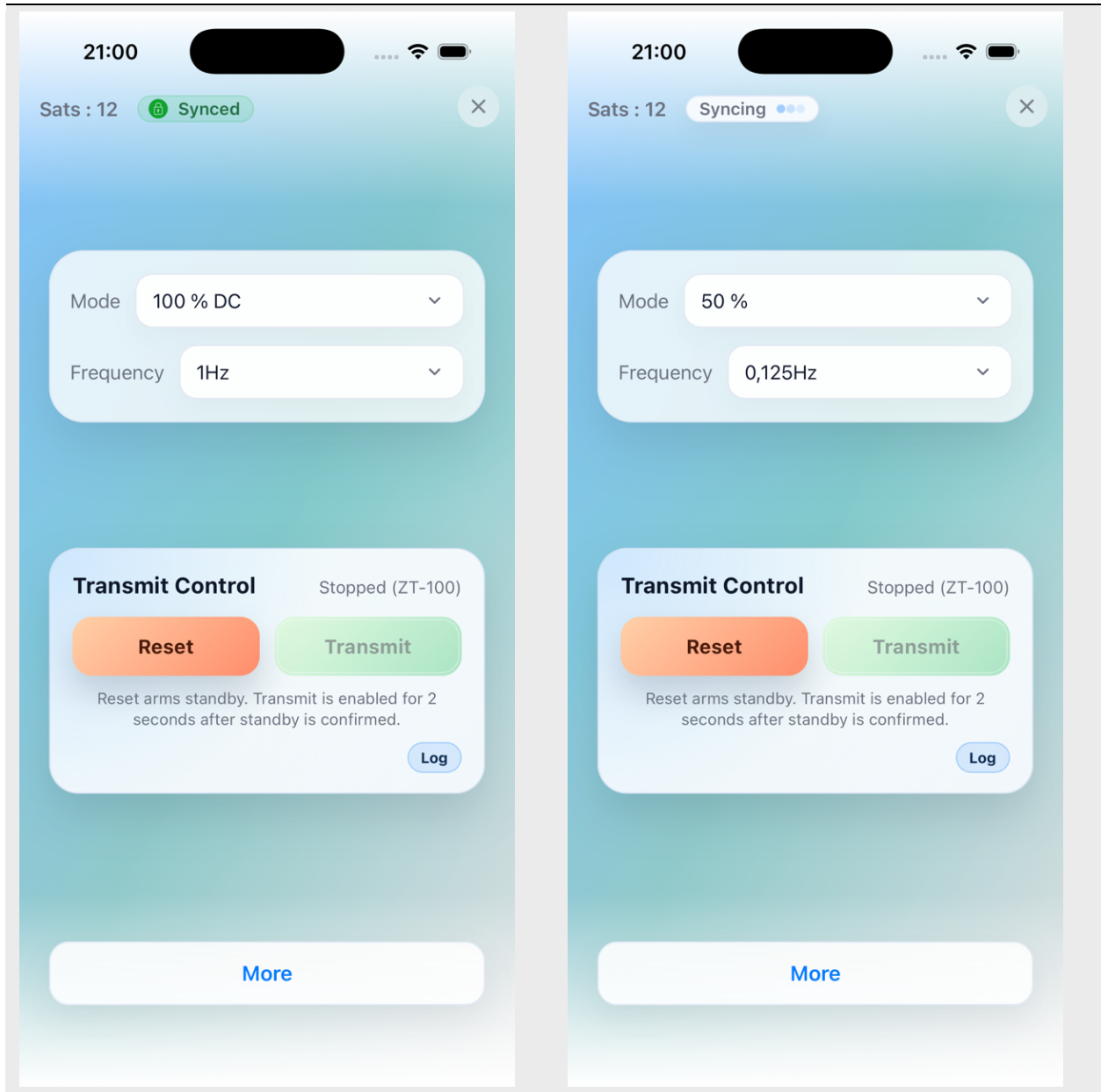
Fig. 2: Current waveform (yellow) and voltage waveform (blue) for a 0.5 Hz 50% duty cycle waveform.

3.7 Stop Transmit

- Press either **RESET** or **TRANSMIT** to stop output.

3.8 Web Interface (Optional)

1. Enable Wi-Fi on the device.
2. In hotspot mode, connect to *ZT-100*.
3. Open `http://10.10.10.10/` in a browser.
4. Or use the mobile app. It automatically connects to the correct network and IP address.



For details, see *Basic Use and Menu Structure* and *Using the Web Interface*.

BASIC USE AND MENU STRUCTURE

4.1 Main Page Overview

The main page provides a quick view of essential information:

- **Selected Mode**
- **Frequency**

16  **RESET** 22°C 

MODE : 100% DC

FREQ : 1 Hz

The top status area shows:

- **Number of satellites** (GPS fix quality)
- **Sync status** (searching vs locked)
- **ZT-100 ID**
- **Wi-Fi status**

4.2 Button Basics

Long-press is about 2 seconds.

- **UP (short press)**: Move up in menus or increase a value.
- **DOWN (short press)**: Move down in menus or decrease a value.
- **SELECT (short press)**: Toggle selection on the main screen or select a menu item.
- **UP/DOWN (long press)**: Adjust Mode on the main screen or make fine changes in the value editor.
- **SELECT (long press)**: Enter/exit menus or confirm a value edit.
- **RESET**: Starts the reset/arm sequence, or stops transmit.
- **TRANSMIT**: Starts transmit during the arm window, or stops transmit.

4.3 Mode Adjustment on Main Page

To switch modes on the main page:

1. Make sure **MODE** is selected (use **SELECT**).
2. Press and hold **UP** or **DOWN** for about 2 seconds.

16 TX 70.3A 22°C

MODE : 50 %

FREQ : 0.125 Hz

4.4 Frequency Adjustment on Main Page

To adjust frequency on the main page:

1. Make sure **FREQUENCY** is selected (use **SELECT**).
2. Use **UP/DOWN** to increase or decrease the frequency.

18 TX 70.8A 25°C

MODE : 50 %

FREQ : 0.25 Hz

4.5 Modes

- **100% DC**: Full frequency range.
- **50%**: Limited frequency range (currently 0.0078125 Hz to 32 Hz).
- **Custom**: User-defined sequence mode (file upload via web app).
- **MMR 5Hz**: Fixed 5 Hz mode (frequency selection hidden).

In **Basic** XMT mode, only **100% DC** and **50%** are available. In **Advanced** XMT mode, all four modes are available. You can switch between **Basic** and **Advanced** either from the device menu (**More Info** -> **XMT Mode**) or from the web interface **More** panel.

4.6 Menu Options

From the MENU screen you can:

- Adjust **Input Under Voltage** and **Input Over Voltage** limits.

- Adjust **Minimum Resistance** and **Maximum Resistance** limits.
- Open **More Info...** for status and system tools.

```

16 TX 70.3A 22°C
-----
Input Under V : 10V
Input Over V : 60V
Min Res : 3.0 Ohm
Max Res : 25.0 Ohm
More Info...

```

4.7 Value Editor

Selecting a limit opens the value editor:

- **UP/DOWN (short press)**: Coarse steps.
- **UP/DOWN (long press)**: Fine steps.
- **SELECT (short press)**: Toggle coarse/fine mode.
- **SELECT (long press)**: Save and return.

```

Frequency
0.125 Hz

```

4.8 More Info Screen

The More Info screen shows status and configuration:

- **XMT Mode**: Toggle Basic/Advanced.
- **GPS Time / GPS Date**
- **Latitude / Longitude / Altitude**
- **GPS Details...**: Opens the GPS details list.
- **TX Log...**: Opens the transmit event log.
- **Temp unit**: C or F.
- **Version**: Firmware version.
- **ZT-100 #**: Instrument ID.
- **Screen Light**: On/Off.
- **Wi-Fi status**: On/Off.

- IP: Shown when Wi-Fi is enabled.

```
16  TX 70.3A  22°C    
Lon  : -105.27  
Alt  : 1630 m  
GPS Details...  
TX Log...  
Temp unit : C
```

4.9 GPS Details Screen

The GPS details list includes:

- HDOP
- Fix age and time age
- Speed and course
- Satellites used
- GPS satellites and GLONASS satellites
- PPS count

```
16  TX 70.3A  22°C    
Fix age : 120ms  
Time age : 200ms  
Speed : 0.0 km/h  
Course : 0.0 deg  
Sats used : 16
```

4.10 Transmit Workflow (Step-by-Step)

1. Press **RESET** to start the reset/arm sequence.
2. The system checks all protection and status conditions.
3. If no errors are reported, the screen shows **RDY** (arm window).
4. Press **TRANSMIT** during the arm window.
5. The system measures load resistance (**MOR**).
6. If the load is in range, transmit begins. If not, **ORE** is shown.
7. Press **TRANSMIT** or **RESET** to stop transmit and return to safe state.

4.11 Status Labels (Top Bar)

The top bar shows a short label for the transmit state. Common labels are:

- **ZT-100**: Boot/initial state.
- **RESET**: Reset sequence active or waiting in safe state.
- **RDY**: Armed window open for transmit.
- **MOR**: Measuring load resistance.
- **ORE**: Load resistance out of range or timed out.
- **TRANSMIT**: Transmit active.
- **STOPPED**: Transmit stopped by operator.
- **ERR**: Shared error line asserted (failsafe wait state).
- **R x.x**: Last measured resistance shown during transmit.
- **TX x.xA**: RMS current shown during transmit.

4.12 Error Codes

Protection codes such as **SM EMER**, **MM OCP**, and **GM OTMP** identify the subsystem and condition that stopped operation. Any error code disables the transmitter output and returns the controller to a safe wait state.

For the meaning of every firmware-defined code and the recommended operator response, see *Error Codes and Corrective Actions*.

USING THE WEB INTERFACE

5.1 Overview

When Wi-Fi is enabled on the ZT-100, the device hosts a web interface that mirrors the front-panel controls and provides administrative tools.

5.2 Mobile App Availability

The app is available on the Apple App Store as **Zonge ZT-100**. An Android version is planned for release soon.

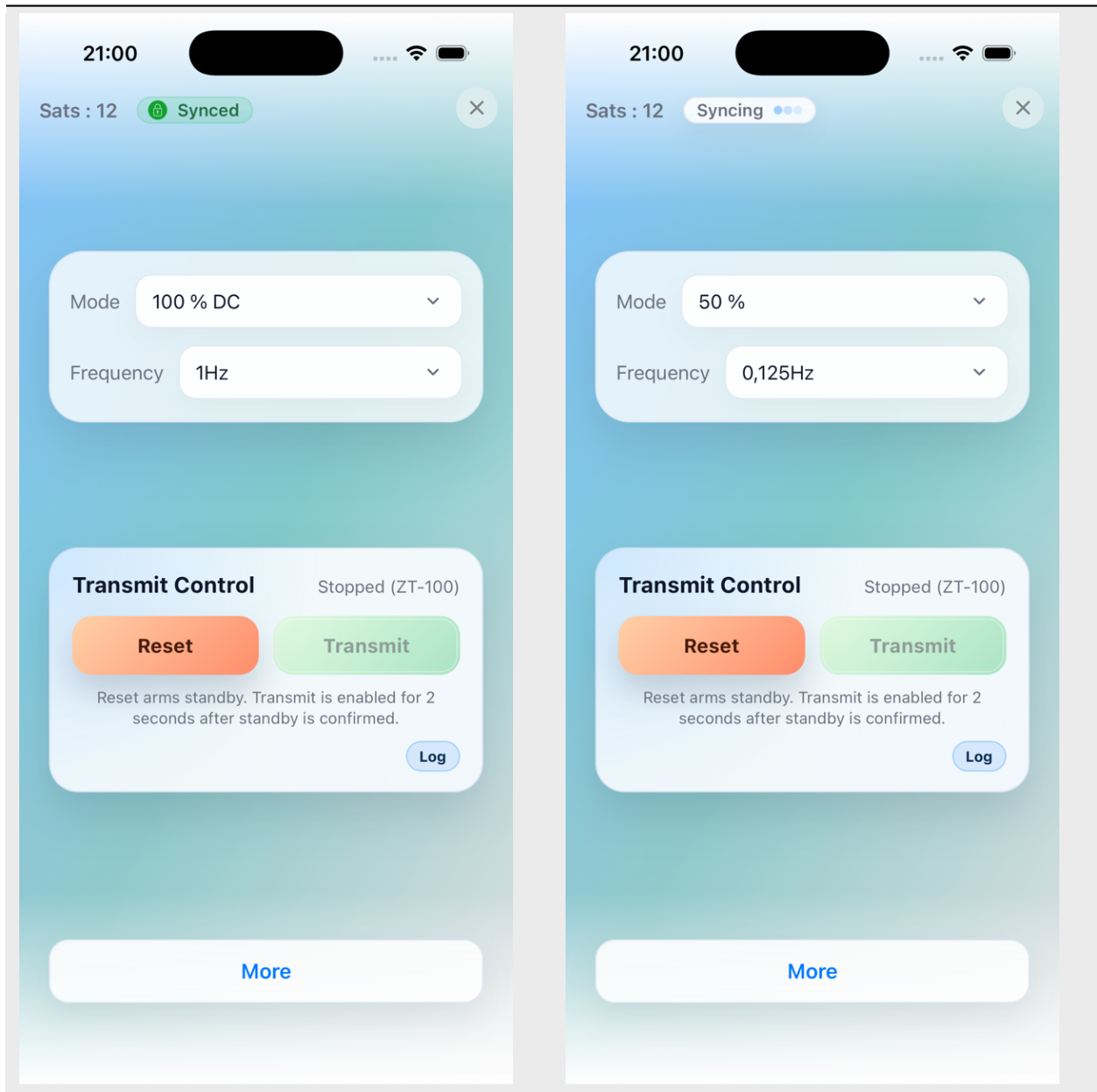
Hotspot mode (default):

- Join the device Wi-Fi network (SSID typically *ZT-100*).
- Open a browser and go to `http://10.10.10.10/`.
- Or use the mobile app, which automatically connects to the correct network and IP address.

If the device is on a local network, use its IP address instead.

5.3 Main Page Controls

The main page shows GPS status, sync status, mode, and frequency. It also provides transmit control and the TX log.



Mode

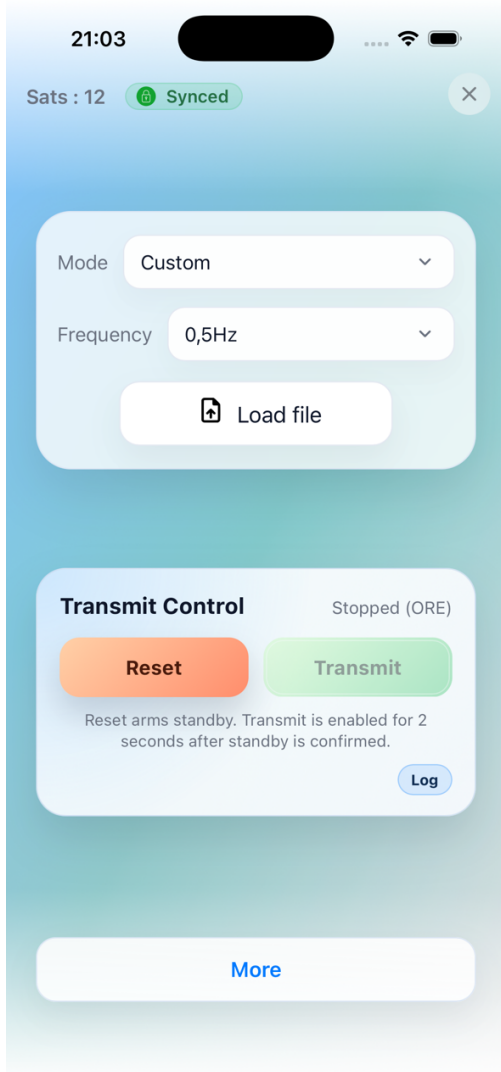
- Select **100% DC**, **50%**, **Custom**, or **MMR 5Hz**.
- In **Basic** XMT mode, only **100% DC** and **50%** appear.
- In **Advanced** XMT mode, all four modes appear.

Frequency

- Sets frequency from 0.0078125 Hz up to 8192 Hz.
- In **50%** mode the maximum is limited (currently 32 Hz).
- In **MMR 5Hz** mode, frequency selection is hidden.

Custom Files (Custom mode)

- A file loader and file list appear only in **Custom** mode.
- Upload **.usm** files. The UI rejects files larger than 256 KB.
- Use the list to select the active custom file.



Transmit Control

- **RESET**: Starts the reset/arm sequence, or stops transmit.
- **TRANSMIT**: Starts transmit during the arm window, or stops transmit.
- **Status line**: Shows *Stopped*, *Standby ready*, *Transmitting*, and includes the current status label in parentheses.

More

- Opens a panel with GPS time, latitude, longitude, altitude, and **XMT Mode** (Basic/Advanced).
- You can switch between **Basic** and **Advanced** here, or from the device screen menu (**More Info -> XMT Mode**).

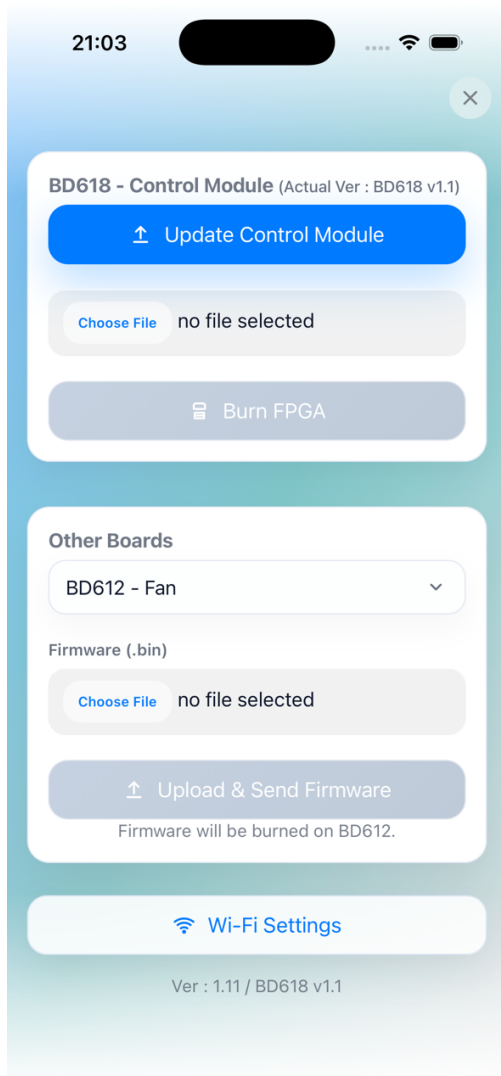
5.4 How to Transmit (Web UI)

1. Click **Reset**.
2. You then have about 2 seconds to click **Transmit**.
3. If everything is fine and load checks pass, transmit starts.
4. Click **Stop** to end transmit.

5.5 Admin Page

Navigate to /admin for firmware and FPGA updates.

- **Update Main Firmware:** Upload the firmware for the main controller.
- **Burn FPGA:** Upload the FPGA image.
- **Other Hardware Targets:** Select the target listed in the UI and upload the matching firmware file.



If the selected target and firmware file do not match, the upload is rejected.

5.6 Wi-Fi Settings Page

Navigate to /wifi from the Admin page.

- **Network Mode:** Hotspot or Local Network.
- **Scan Networks:** Populate the available SSID list.
- **Connect:** Apply the selected mode and credentials.

21:04

Back

Network Mode

Local Network

Current Saved Network

n/a

Available Networks

Scan Networks

Home Sweet Home

Local Wifi Password

Connect

Ver : 1.11 / BD618 v1.1

GENERATE USER DEFINED SEQUENCES

In addition to the pre-programmed 50% and 100% duty-cycle control signals the ZT-100 allows user-defined control signals in **Custom** mode. This enables specialized ternary transmission sequences when required by the survey design.

6.1 Control Signals

The transmitter output can have three states:

1. Positive ON
2. OFF
3. Negative ON

Positive ON and Negative ON refer to current flow through the transmitter in opposite directions. The ZT-100 defines the output signal using two binary control signals:

1. ON#
2. POL

Each control signal can be LOW or HIGH. HIGH corresponds to 3.3 V and LOW corresponds to 0 V.

- **ON#** controls whether the transmitter is on. The # indicates **active low**. When ON# is LOW the transmitter is on; when ON# is HIGH the transmitter is off.
- **POL** controls the direction of current flow. POL HIGH corresponds to positive current flow, POL LOW corresponds to negative current flow.

6.2 Output Signal Example

The red graph shows ON#, the green graph shows POL, and the black graph shows resulting transmitter output. The sequence includes all three states.

Note

The control signals are binary and do not contain amplitude information. The output signal amplitude is set on the transmitter hardware.

6.3 Generating the Signal Definition File

To use a user-defined signal, create a **.usm** file and upload it via the web interface. The file contains a sequence of control signal states. At each base clock tick, the next POL and ON# values are read and applied. When the end of file is reached the sequence loops back to the beginning.

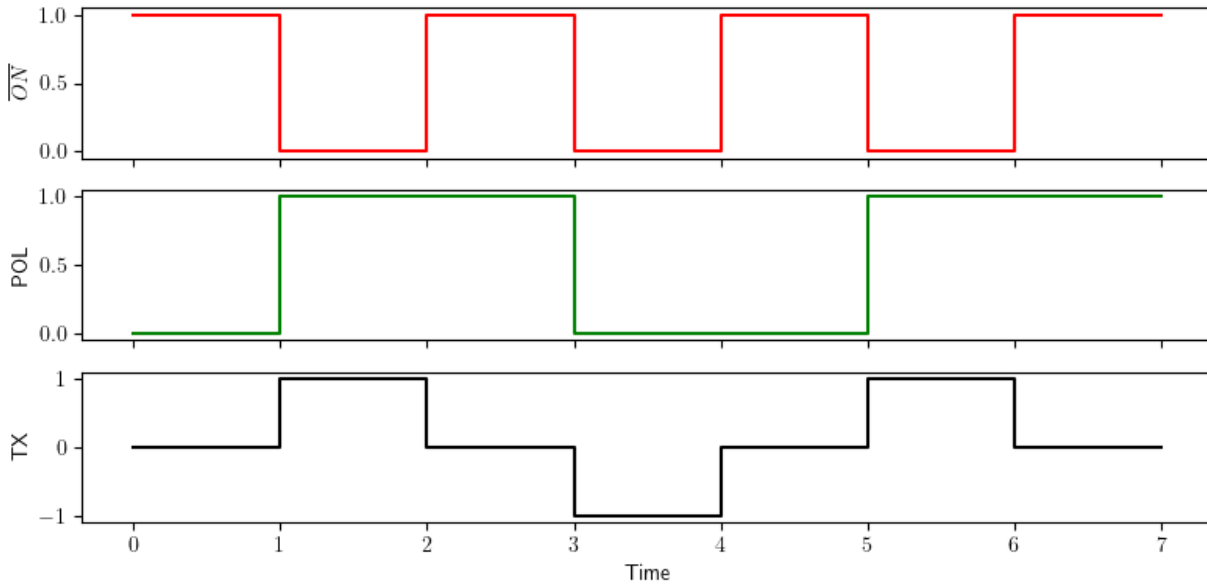


Fig. 1: Generation of a 50% duty-cycle output signal.

6.4 File Format

The .usm file is stored in binary form and contains three parts:

1. Sequence length
2. List of POL entries
3. List of ON# entries

The sequence length is a 16-bit unsigned integer in big-endian byte order, allowing up to 65,536 entries.

POL and ON# entries are stored as packed bits. Every eight consecutive bits are stored as a byte. If the sequence length is not a multiple of 8, pad the final byte with zeros. The firmware ignores padded bits based on the length value.

6.5 Example: Generate a PRBS .usm File

The following Python code generates a 4th order PRBS sequence file:

```

1  import numpy as np
2  np.set_printoptions(formatter={'int':hex})
3
4  pol = [1,1,0,0,0,1,0,0,1,1,0,1,0,1,1]           # sequence for polarity
5  on_ = [0,0,0,0,0,0,0,0,0,0,0,0,0,0,0]           # sequence for ON# (active low)
6  slen = len(pol)                                    # sequence length
7
8  # Pad to a multiple of 8
9  apol = 8-len(pol)%8
10 for i in range(apol):
11     pol.append(0)
12     on_.append(0)
13

```

(continues on next page)

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```

14 pol_bytes = (np.packbits(np.array(pol),bitorder='big'))
15 on_bytes  = (np.packbits(np.array(on_),bitorder='big'))
16 print(pol_bytes)
17 print(on_bytes)
18
19 print('Sequence length = {}, {}'.format(slen,slen.to_bytes(2,byteorder='big')))
20 print("Pol array = {}".format(pol))
21 print("ON_ array = {}".format(on_))
22
23 with open ("PRBS_4.usm", "wb") as binary_file:
24     binary_file.write(slen.to_bytes(2,byteorder='big'))
25     binary_file.write(pol_bytes)
26     binary_file.write(on_bytes)

```

6.6 Uploading a .usm File

1. Enable Wi-Fi on the ZT-100.
2. Open the web interface and switch to **Custom** mode.
3. Use the file loader to upload the .usm file.

6.7 Transmitting a User Defined Sequence

The .usm file defines the control sequence only. Select the desired base frequency in the web interface to generate the final transmitter waveform.

TROUBLESHOOTING

This section focuses on the most common operational issues and how to resolve or diagnose them using the screen and web interface.

7.1 No GPS Lock / Sync Searching

Symptoms:

- Satellite count is low or zero.
- Sync indicator shows searching.

Actions:

- Verify the GPS antenna is connected to the BNC port.
- Move to an open area with clear sky view.
- Wait a few minutes for initial acquisition.

7.2 Transmit Button Disabled

Symptoms:

- **Transmit** is disabled on the web UI.
- The screen does not show **RDY**.

Actions:

- Press **RESET** to start the reset/arm sequence.
- Wait for **RDY** (arm window). The window is short (about 2 seconds).
- If **RDY** never appears, note the displayed code and use the *Error Codes and Corrective Actions*.

7.3 Error Codes and Corrective Actions

An error code identifies the protection function or subsystem that prevented operation. The controller disables the transmitter output and returns to a safe wait state whenever an error is detected.

Codes beginning with a subsystem prefix identify the reporting module:

- **SM** – Safety module
- **MM** – Monitoring module
- **GM** – Gate-driver module

- **FM** – Cooling/fan module

Warning

An error code means that a protection condition has stopped operation. Do not bypass an interlock or repeatedly reset the unit without correcting the cause. Connected power equipment may remain hazardous even though the ZT-100 has disabled its transmitter output.

Table 1: Error-code reference

Display code	Meaning	Operator response
ERR	The shared system error line is active, but a more specific source has not been identified.	Keep the unit stopped. Check the supply, transmitter, load, and cable connections. After correcting any visible problem, press RESET once to run the protection check again. If ERR returns without a more specific code, contact Zonge International technical support.
ORE	The measured load resistance is outside the configured Minimum Resistance and Maximum Resistance limits, or the resistance measurement did not complete within the allowed time.	Check the transmitter cable and load circuit, then confirm that both resistance limits are appropriate for the installation. Press RESET and repeat the transmit sequence only after correcting the cause.
SM EMER	The emergency-stop input is active.	Make the work area safe, determine why the emergency stop was activated, and release it only when safe to do so. Verify the emergency-stop circuit, then press RESET .
SM GIUV	The safety module's global input under-voltage protection is active.	Verify the input power source and power connections. Restore the input voltage to the permitted range, then press RESET .
SM GIOV	The safety module's global input over-voltage protection is active.	Verify the input power source and power connections. Restore the input voltage to the permitted range, then press RESET .
SM UDIUV	The input voltage is below the user-defined Input Under Voltage limit.	Compare the configured limit with the actual input voltage. Correct the supply condition or, if authorized, set an appropriate limit; then press RESET .
SM UDIOV	The input voltage is above the user-defined Input Over Voltage limit.	Compare the configured limit with the actual input voltage. Correct the supply condition or, if authorized, set an appropriate limit; then press RESET .
SM LERR	The safety module has asserted its local error flag without a more specific safety input code.	Confirm that the emergency-stop and input-voltage conditions are normal, then press RESET once. If the code returns, contact Zonge International technical support.
SM ERR	The safety module reported an error that is not represented by a more specific display code.	Record the code and operating conditions. Press RESET only after all safety inputs are normal. If the code returns, contact Zonge International technical support.
MM OVP	Over-voltage was detected on the positive output half-cycle.	Keep the unit stopped and inspect the transmitter, load, cable, and operating setup. Press RESET only after correcting the cause. If the code returns, have the system serviced.
MM OVN	Over-voltage was detected on the negative output half-cycle.	Keep the unit stopped and inspect the transmitter, load, cable, and operating setup. Press RESET only after correcting the cause. If the code returns, have the system serviced.
MM OCP	Over-current was detected on the positive output half-cycle.	Keep the unit stopped and inspect the transmitter, load, cable, and operating setup. Press RESET only after correcting the cause. If the code returns, have the system serviced.
MM OCN	Over-current was detected on the negative output half-cycle.	Keep the unit stopped and inspect the transmitter, load, cable, and operating setup. Press RESET only after correcting the cause. If the code returns, have the system serviced.
MM LERR	The monitoring module has asserted its local error flag without a more specific voltage or current code.	Press RESET once after confirming that the transmitter and load connections are correct. If the code returns, contact Zonge International technical support.
MM ERR	The monitoring module reported an error that is not represented by a more specific display code.	Record the code and operating conditions. Press RESET only after checking the transmitter and load. If the code returns, contact Zonge International technical support.
GM OTMP	The gate-driver module temperature exceeded its protection threshold.	Allow the unit to cool. Check ambient temperature, ventilation, and fan operation then press RESET . If the

7.3. Error Codes and Corrective Actions

The display reports one protection label at a time. If more than one condition is active, correcting the displayed condition and pressing **RESET** may reveal another code. The unit is ready to transmit only when the reset check completes and **RDY** appears.

If a code persists after the stated checks, record the exact code, operating mode, frequency, connected equipment, and the conditions immediately before the stop. Provide this information to Zonge International technical support.

7.4 Wi-Fi Not Visible

Symptoms:

- No ZT-100 SSID appears.

Actions:

- On the front panel, open **More Info** and set **Wi-Fi status** to **ON**.
- Wait for the Wi-Fi icon to appear on the top bar.

7.5 Can't Open the Web UI

Symptoms:

- Browser cannot reach the device page.

Actions:

- If in hotspot mode, connect to the device SSID and open `http://10.10.10.10/`.
- If in local network mode, use the IP address shown on the screen.
- Confirm **Wi-Fi status** is **ON**.

7.6 Custom File Upload Fails

Symptoms:

- Upload rejected or file not listed.

Actions:

- Ensure the file extension is **.usm**.
- The web UI rejects files larger than 256 KB.
- Stay in **Custom** mode while uploading.

7.7 OTA Update Fails

Symptoms:

- Upload rejected or target mismatch.

Actions:

- Confirm the firmware file matches the selected target in the web UI.
- If the upload is rejected, re-check target selection and file version.
- Keep the browser open until the update process completes.

SPECIFICATIONS

CONNECTOR TYPES AND PINOUTS.

9.1 External Transmitter Control

The transmitter-control connector is a **15-pin female** MIL-spec circular connector. This allows compatibility with older generation of Zonge equipment.

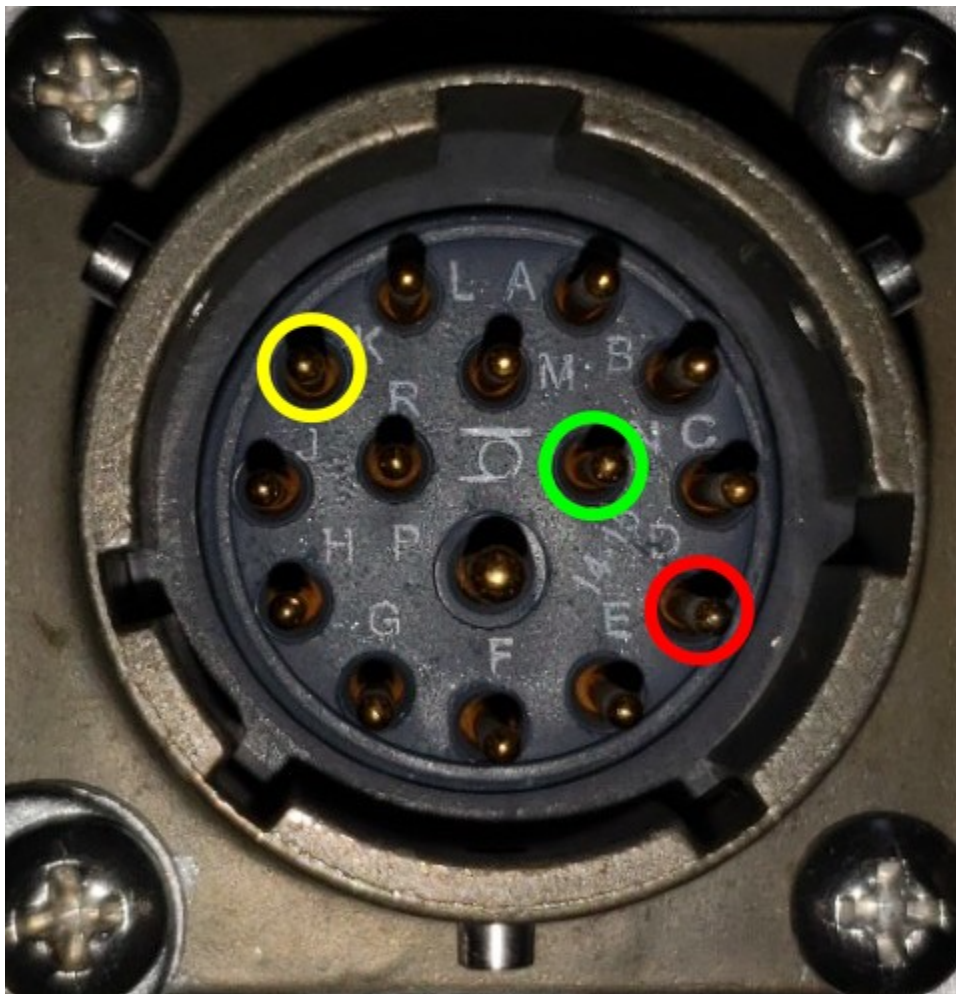


Fig. 1: Pinout of the 15-pin MIL transmitter controller connector. POL is on pin D (red), COMMON (GND) is on pin N (green) and ON/OFF is on pin K (yellow).

Use Zonge cable type **XMT/16-CN/6** for compatible transmitter/control connections.

9.1.1 GPS Antenna Connector

The GPS antenna connects via a 50 Ohm BNC connector. Use a 3.3 V active GPS antenna with minimum gain of 15 dB and maximum gain of 30 dB.

INTRODUCTION

Warning



The ZT100-B is a high power transmitter for geophysical purposes. It operates at lethal voltages. Only trained, qualified personnel should operate, maintain or service it.

This manual covers day-to-day operation of the ZT100-B control system.

The ZT-100 provides GPS-synchronized transmitter control with:

- Preset **100% DC** and **50%** operating modes
- Configurable frequency selection across a wide range
- Support for user-defined control signals in **Custom** mode
- Status, diagnostics, and firmware management through the device interface

10.1 User Interfaces

You can operate the ZT-100 in two ways:

1. **Front Panel:** Local screen-and-button control for mode, frequency, and core status functions.
2. **Web Interface:** Wi-Fi browser access for control, logs, updates, and network settings.

DISPLAY MESSAGES

GIUV: General Input Under Voltage